

Supplementary Summer Examination – 2025

Course: B.Tech.

Branch: Civil Engineering

Semester: VI

Subject Code & Name: BTCVC601 & Design of RC Structures

Max Marks: 60

Date:30/06/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.
7. IS 456-2000, IS 875-PART-1,2,3 are allowed.

		CO	Marks
Q. 1 Objective type questions. (Compulsory Question)			12
1 In the Working Stress Method (WSM), the design is based on:		CO1	1
a. Collapse load of the structure	b. Yield strength of the material	c. Elastic behavior under service loads	d. Ultimate limit state behavior
2 In the Limit State Method, the limit state of collapse deals with		CO1	1
a. Excessive deflection	b. Durability issues	c. Structural failure due to strength inadequacy	d. Vibration of floors
3 Which section is most ductile?		CO2	1
a. Balanced section	b. Under-reinforced	c. Over-reinforced	d. All are equally ductile
4 The maximum permissible stress in Fe 415 steel in Working Stress Method (WSM) is:		CO2	1
a. 230 MPa	b. 250 MPa	c. 280 MPa	d. 415 MPa
5 In the strain variation diagram for a flexural member, the strain distribution is:		CO2	1
a. Parabolic	b. Uniform	c. Linear	d. Logarithmic
6 Which factor ensures reliability by covering variability in material properties and workmanship?		CO3	1
a. Load factor	b. Factor of rigidity	c. Partial safety factor	d. Strain factor
7 Which of the following does not affect bond strength between steel and concrete?		CO3	1
a. Bar diameter	b. Concrete grade	c. Bar spacing	d. Bar surface condition

8	Bond failure is more likely to occur in:	CO3	1
	a. short bars under compression b. long bars in tension c. Central part of a simply supported beam d. in supports under compression.		
9	Doubly reinforced beams are preferred when:	CO2	1
	a. Load is very light b. Beam depth is restricted c. Only shear acts d. Simply supported span is small		
10	In a doubly reinforced section, compression steel is considered effective if:	CO2	1
	a. It lies outside the concrete b. It is not enclosed by stirrups c. It is provided with proper development length d. It is in tension zone		
11	The axial design load capacity of a short column (as per LSM) is	CO2	1
	a. $0.67f_{ck}A_c + 0.87f_yA_{sc}$ b. $0.45f_{ck}A_c + 0.75f_yA_{sc}$ c. $0.4f_{ck}A_c + 0.67f_yA_s$ d. $0.4f_{ck}A_c + 0.67f_yA_{sc}$		
12	Which condition represents the pure axial load in an interaction diagram?	CO2	1
	a. At zero moment b. At maximum eccentricity c. At neutral axis passing through tension face d. At zero axial load		
Q. 2 Solve the following.			12
A)	Explain the term 'Limit State of Collapse' and state why serviceability is an important consideration in structural design?	CO1	6
B)	Explain the application of Partial Safety Factors (PSFs) in the Limit State Method (LSM) for both materials and loads. Provide typical values for concrete, steel, dead load, and live load as per IS codes.	CO1	6
Q.3 Solve the following.			12
A)	A singly reinforced rectangular concrete beam is 250 mm wide and 500 mm deep. The beam is reinforced with 3 bars of 20 mm diameter. Assume Safe stress in concrete = 7 N/mm ² , safe stress in steel = 230 N/mm ² , Modular ratio = 13.33. Using Working Stress Method, calculate the moment of resistance of the section.	CO2	6
B)	Define the term "stress block" and explain its parameters as per IS 456:2000.	CO3	6
Q. 4 Solve Any Two of the following.			12
A)	What is the Limit State of Collapse in Shear? Explain types of shear reinforcement.	CO3	6
B)	RCC beam is 250 mm wide and 450 mm deep. It uses Fe 415 steel. Design minimum shear reinforcement per IS 456:2000. Use 8 mm 2-legged stirrups.	CO2	6
C)	Define bond stress. Derive the expression for development length of a reinforcement bar as per IS 456:2000.	CO3	6

Q.5	Solve Any Two of the following.		12
A)	A one-way slab spans 3 m. Assume 120 mm thickness, LL = 3 kN/m ² , FF = 1 kN/m ² . Design main reinforcement. Use M20 and Fe 415.	CO1	6
B)	Design a dog-legged staircase for floor height = 3 m, tread = 250 mm, rise = 150 mm. Assume slab thickness = 150 mm, LL = 3 kN/m ² .	CO3	6
C)	Explain the behavior of flanged (T and L) beams with their design considerations.	CO1	6
Q. 6	Solve Any Two of the following.		12
A)	What are axially loaded short columns? Explain the design steps as per IS 456:2000.	CO3	6
B)	Design a rectangular short column 300 × 500 mm subjected to 1200 kN axial factored load using M25 and Fe 500.	CO1	6
C)	What is eccentricity? Describe minimum eccentricity and its effect on column design.	CO3	6

*** End ***